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ELECTRO-HYDROSTATIC ACTUATOR

Abstract

Embodiments of the present disclosure provide a method and apparatus having an actuator assembly. An exemplary actuator assembly includes a cylinder body having a longitudinal axis and a plurality of passageways, and an accumulator fluidly connected to the plurality of passageways. The exemplary actuator assembly also includes a solenoid assembly comprising a core tube, an armature, a first magnet, a second magnet, and an excitation coil, the core tube extending through the longitudinal axis, the core tube having a first ferromagnetic end section and a spaced apart second ferromagnetic end section.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present disclosure relates to an electro-hydrostatic actuator and more particularly to an electro-hydrostatic actuator pump.

Description of Related Art

[0002] Electro-hydrostatic actuators are hydraulic systems with self-contained actuators that operate solely by electrical power. Electro-hydrostatic actuators eliminate the need to separate hydraulic pumps and tubing. An electro-hydrostatic actuator having a self-contained design can help improve safety and reliability of the systems.

BRIEF SUMMARY OF THE INVENTION

[0003] In view of the foregoing, it is an object of the present disclosure to provide a method and apparatus.

[0004] A first exemplary embodiment of the present disclosure provides an actuator assembly including cylinder body having a longitudinal axis and a plurality of passageways, and an accumulator fluidly connected to the plurality of passageways, the accumulator operable to maintain a fluid under pressure and to urge the fluid through the plurality of passageways. The actuator assembly further includes a moveable cylinder fluidly connected to the plurality of passageways, the moveable cylinder operable to move through the longitudinal axis in response to pressure from the fluid. The actuator assembly still further includes a solenoid assembly comprising a core tube, an armature, a first magnet, a second magnet, and an excitation coil, the core tube extending through the longitudinal axis, the core tube having a first ferromagnetic end section and a spaced apart second ferromagnetic end section, the armature having a rod extending through the longitudinal axis, the armature disposed longitudinally between the first ferromagnetic end section and the spaced apart second ferromagnetic end section, the first magnet and the second magnet disposed radially outside the core tube, the first magnet separated from the second magnet by a ferromagnetic spacer, the excitation coil located radially outward from the first and second magnets

[0005] A second exemplary embodiment of the present disclosure provides a method. The method includes providing a cylinder body having a longitudinal axis and a plurality of passageways, and providing an accumulator fluidly connected to the plurality of passageways, the accumulator operable to maintain a fluid under pressure and to urge the fluid through the plurality of passageways. The method further includes providing a moveable cylinder fluidly connected to the plurality of passageways, the moveable cylinder operable to move through the longitudinal axis in response to pressure from the fluid. The method still further includes providing a solenoid assembly comprising a core tube, an armature, a first magnet, a second magnet, and an excitation coil, the core tube extending through the longitudinal axis, the core tube having a first ferromagnetic end section and a spaced apart second ferromagnetic end section, the armature having a rod extending through the longitudinal axis, the armature disposed longitudinally between the first ferromagnetic end section and the spaced apart second ferromagnetic end section, the first magnet and the second magnet disposed radially outside the core tube, the first magnet separated from the second magnet by a ferromagnetic spacer, the excitation coil located radially outward from the first and second magnets.

[0006] A third exemplary embodiment of the present disclosure provides an apparatus. The

apparatus includes a fixed body having a first side and an opposite second side, the fixed body defining at least one opening, the at least one opening providing a passageway between the first side and the opposite second side. The apparatus further includes a flexible body having a flexible first side and an opposite flexible second side, the flexible body having at least one flexible portion and a spaced apart stationary portion, the at least one flexible portion operable to flex with respect to the stationary portion such that at least one passageway from the flexible first side to the opposite flexible second side is formed when the at least one flexible portion is in a flexed position.

[0007] A fourth exemplary embodiment of the present disclosure provides an actuator assembly. The actuator assembly includes a cylinder body having a plurality of passageways, and a first accumulator fluidly connected to the plurality of passageways, the first accumulator operable to maintain a fluid under pressure and to urge the fluid through the plurality of passageways. The actuator assembly further includes a second accumulator fluidly connected to the plurality of passageways, the second accumulator operable to maintain a fluid under pressure and to urge the fluid through the plurality of passageways, the second accumulator fluidly connected to the first accumulator through the plurality of passageways and operable to receive fluid from the first accumulator, and a moveable cylinder fluidly connected to the plurality of passageways, the moveable cylinder operable to move in response to pressure from the fluid. The actuator assembly still further includes a solenoid assembly comprising a core tube, an armature, a first magnet, a second magnet, and an excitation coil, the core tube extending through a longitudinal axis, the core tube having a first ferromagnetic end section and a spaced apart second ferromagnetic end section, the armature having a rod extending through the longitudinal axis, the armature disposed longitudinally between the first ferromagnetic end section and the spaced apart second ferromagnetic end section, the first magnet and the second magnet disposed radially outside the core tube, the first magnet separated from the second magnet by a ferromagnetic spacer, the excitation coil located radially outward from the first and second magnets.

[0008] The following will describe embodiments of the present disclosure, but it should be appreciated that the present disclosure is not limited to the described embodiments and various modifications of the disclosure are possible without departing from the basic principle. The scope of the present disclosure is therefore to be determined solely by the appended claims.

[0009] BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0010] FIG. 1 presents a perspective view of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0011] FIG. 2 presents a cross-sectional view of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0012] FIG. 3 presents a cross-sectional view of an embodiment of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0013] FIG. 4 presents a block diagram of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0014] FIG. 5 presents a perspective interior view of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0015] FIG. 6 presents another cross-sectional of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0016] FIG. 7 presents a close-up cross-sectional view of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0017] FIG. 8a presents a close-up cross-sectional view of an exemplary shuttle valve suitable for use in practicing exemplary embodiments of this disclosure.

[0018] FIG. 8b presents a close-up cross-sectional view of a portion of an exemplary reed valve suitable for use in practicing exemplary embodiments of this disclosure.

[0019] FIG. 9 presents a cross-sectional view of an alternative exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0020] FIG. **10** presents another cross-sectional view of the alternative exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0021] FIG. **11** presents yet another cross-sectional view of the alternative exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0022] FIG. **12** presents a perspective interior view of the alternative exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0023] FIG. **13** presents a block diagram of the alternative exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0024] FIG. **14** presents another block diagram of the alternative exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0025] FIG. **15** presents yet another block diagram of the alternative exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0026] FIG. **16** presents another cross-sectional view of the alternative exemplary device illustrating flow paths suitable for use in practicing exemplary embodiments of this disclosure.

[0027] FIG. **17** presents another cross-sectional view of the alternative exemplary device illustrating further flow paths suitable for use in practicing exemplary embodiments of this disclosure.

[0028] FIG. **18** presents another cross-sectional view of the alternative exemplary device illustrating further flow paths suitable for use in practicing exemplary embodiments of this disclosure.

[0029] FIG. **19** presents another cross-sectional view of the alternative exemplary device illustrating yet further flow paths suitable for use in practicing exemplary embodiments of this disclosure.

[0030] FIG. **20** presents a block diagram of another exemplary device illustrating flow paths suitable for use in practicing exemplary embodiments of this disclosure.

[0031] FIG. **21** presents a cross-sectional view of an exemplary device illustrating bearings suitable for use in practicing exemplary embodiments of this disclosure.

[0032] FIG. **22** presents a block diagram of yet another exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0033] FIG. **23** presents a block diagram of a further exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0034] FIG. **24** presents a block diagram of a further alternative exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0035] FIG. **25** presents a block diagram of yet a further alternative exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0036] FIG. **26** presents a block diagram of an even further alternative exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0037] FIG. **27** presents a block diagram of another alternative exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0038] FIG. **28** presents a block diagram of yet another exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0039] FIG. **29** presents another block diagram of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0040] FIG. **30** presents yet another alternative block diagram of an exemplary device suitable for use in practicing exemplary embodiments of this disclosure.

[0041] FIG. **31** presents a block diagram of an exemplary device having two accumulators suitable for use in practicing exemplary embodiments of this disclosure.

[0042] FIG. **32** presents another block diagram of an exemplary device having two accumulators suitable for use in practicing exemplary embodiments of this disclosure.

[0043] FIG. **33** presents a cross-sectional view of an exemplary device having two accumulators

suitable for use in practicing exemplary embodiments of this disclosure.

[0044] FIG. **34** presents a cross-sectional view of an exemplary device having two accumulators suitable for use in practicing exemplary embodiments of this disclosure.

[0045] FIG. **35** presents a close-up view of a portion of an exemplary device having two accumulators suitable for use in practicing exemplary embodiments of this disclosure.

[0046] FIG. **36** presents a cross-sectional view of an exemplary reed valve suitable for use in practicing exemplary embodiments of this disclosure.

[0047] FIG. **37** presents a side perspective view of an exemplary reed valve suitable for use in practicing exemplary embodiments of this disclosure.

[0048] FIG. **38** presents a front view of another exemplary reed valve suitable for use in practicing exemplary embodiments of this disclosure.

[0049] FIG. **39** presents an interior view of an exemplary reed valve suitable for use in practicing exemplary embodiments of this disclosure.

[0050] FIG. **40** presents a cross-sectional view of an exemplary reed valve suitable for use in practicing exemplary embodiments of this disclosure.

[0051] FIG. **41** presents a number of alternative embodiments of exemplary reed valve configurations suitable for use in practicing exemplary embodiments of this disclosure.

Description

DETAILED DESCRIPTION OF THE INVENTION

[0052] Traditional electro-hydrostatic actuator systems are large in size. They have less stiffness and lower force output in comparison to their overall size. As such, there is a need for an electro-hydrostatic actuator that is smaller in size and can be operated under low electrical power levels to produce high actuation forces.

[0053] Embodiments of the present disclosure provide an electro-hydrostatic actuator assembly having a single acting cylinder with a spring return. Embodiments include an electro-hydrostatic actuator assembly having an accumulator and hydraulic actuation that is self-contained within a single assembly. Embodiments include an electro-hydrostatic actuator assembly having a solenoid with an embedded magnet design that is operable to pulse to pump oil from the accumulator into the cylinder through a set of pistons and check valves. Embodiments provide the embedded magnet actuator of the electro-hydrostatic actuator assembly can pulse in a first direction to allow for incremental stroke changes of the actuator that are proportional to the magnetic flux created by the embedded magnet actuator. Embodiments provide that the embedded magnet actuator can be energized in a second direction to actuate the drain valve thereby draining the cylinder. The electro-hydrostatic actuator assembly includes a spring in the cylinder operable to force or push oil back to the low-pressure accumulator. In some embodiments, the electro-hydrostatic actuator assembly includes a position sensor that can provide feedback position data of the cylinder to a controller. Embodiments of the electro-hydrostatic actuator assembly uses silicone oil to substantially maintain the viscosity as constant as possible over a wide temperature range in the actuator assembly. Embodiments include using other types of oils that allow the system to operate as set forth below throughout a wide temperature range. Embodiments provide an electro-hydrostatic actuator that consumes a low amount of power when the actuator is in a static position. Embodiments provide an electro-hydrostatic actuator that is turned off or does not consume power when it is not causing or pumping a fluid to flow within the system.

[0054] Referring to FIG. **1**, shown is electro-hydrostatic assembly **100** having a cylinder body **102**, a single acting cylinder **104**, a fill port **106**, and a bleed screw **108**. Cylinder body **102** includes a longitudinal axis **110**. In one embodiment, the cylinder body **102** is approximately 2.625-inch dia by 8.42 inches long. It should be appreciated that embodiments of cylinder body **102** range from

1.5 inch dia by 5 inches long to 4.0 inches dia by 15 inches long.

[0055] Reference is now made to FIGS. 2-4, which depict cross-sectional views of electro-hydrostatic assembly **100**. Shown in FIGS. 2-4 are electro-hydrostatic assembly **100** having a cylinder body **102**. Within cylinder body **102** is a single acting cylinder **104** and a fill port **106**. Cylinder body **102** has a longitudinal axis as indicated by line **110**. Electro-hydrostatic assembly **100** also includes an accumulator **112**, a check valve **120a**, an embedded magnet actuator **116**, a drain valve **118** comprising a spool **122** with a cavity **124** in series with check valve **120b**, a pump piston **126**, and drain path **128** from drain valve **118** to accumulator reservoir **150**.

[0056] Embedded magnet actuator **116** includes a coil **130** (shown in FIGS. 2, 3 and 6) that circumscribes and is radially outward from magnets **132** and armature **134** (shown in FIG. 3). It should be appreciated that embodiments of embedded magnet actuator **116** includes at least one magnet **132** and can include multiple magnets **132** provided it causes embedded magnet actuator **116** to operate as described herein. Coil **130** is operable to have a current passed through it to create a magnetic flux. In some embodiments, coil **130** is made of copper or aluminum insulated magnet wire. Armature **134** is located within cavity **136** (shown in FIG. 3) and is operable to move through the longitudinal axis **110** of cavity **136**. It should be appreciated that embodiments of embedded magnet actuator **116** can include one or more coils **130**. The magnetic flux created by current passing through coil **130** causes armature **134** to move through the longitudinal axis **110** of cavity **136**. The direction of movement of armature **134** is determined by the relationship with the pole faces, the orientation of magnets **132**, and the polarity of the current applied to coil **130**. Armature **134** and cavity **136** are maintained within a core tube **138** (shown in FIG. 6).

[0057] Radially outward from the core tube **138** are spaced apart magnets **132**. Spaced apart magnets **132** circumscribe the core tube **138** and may each include two or more separate (e.g., C-shaped or curved) magnets. Magnets **132** are separated or spaced from one another in the longitudinal direction by a ferromagnetic spacer **133** (shown in FIGS. 3 and 6) that circumscribes the core tube **138**. It should be appreciated that embodiments of ferromagnetic spacer **133** can include two or more separate spacers (e.g., curved spacers) that collectively circumscribe the core tube **138**. Magnets **132** are positioned such that like poles face one another.

[0058] Armature **134** includes or is affixed to rod **140**. Rod **140** extends through the longitudinal axis **110** and is operable to move in the longitudinal direction with armature **134** in response to changes in magnetic flux caused by current passing through coil **130** in a first polarity direction and a second polarity direction. Rod **140** is operable to move through the longitudinal direction within core tube **138** along flexible bearings **186** (shown in FIG. 21) located on the longitudinal exterior face of pole stops **188**, **190**. Rod **140** is operable to put pressure on, compress and move piston **126** from a first neutral position to a second compressed position. This movement creates a suction force within piston **126** causing the check valve **120c** to open allowing fluid to flow from the accumulator reservoir **150** into the pumping cavity when rod **140** and armature **134** is moved towards the accumulator **112**. Upon removal of magnetic flux, the armature **134**, rod **140**, and piston **126** move away from the accumulator **112** towards their neutral position or centered position within core tube **138** located radially inward from magnets **132**. The check valve **120c** will close and the force created by spring **142** (shown in FIG. 3) pushes the piston **126** with the check valve **120c** in a closed position forcing fluid at a pressure equal to the force created by spring **142** divided by the area of the piston **126** through a second series of check valves **120d** into the actuator cavity **146c**.

[0059] Rod **140** is also operable to put pressure on and move check valve **120b** followed by spool **122** from the closed or neutral position obstructing a flow of fluid to the open position allowing the flow of fluid from the cylinder cavity **146c** back to the accumulator reservoir **150**. This occurs when rod **140** and armature **134** is moved towards single acting cylinder **104** in response to a second polarity (e.g., a polarity that is opposite the first polarity) signal applied to coil **130**. Upon removal of the magnetic field from coil **130**, rod **140**, armature **134**, and piston **122** move away

from the actuator **104** by spring **144** back to their neutral positions obstructing fluid from the actuator cavity **146c** to accumulator reservoir **150**. Check valve **120b** then closes blocking fluid from exiting actuator cavity **146c**. It should be appreciated the embodiments include embedded magnet actuator **116** being replaced by any type of bi-directional electrical actuator that is operable to compress or activate two piston pumps independently as described herein. Embodiments of bi-directional electrical actuator include voice coil actuators, moving magnet actuators, and one or more motors operable to actuate lateral movement of a rod that can independently compress or activate two piston pumps or via a cam to independently compress or activate the two piston pumps.

[0060] Accumulator **112** is fluidly connected to single acting cylinder **104** within cylinder body **102** through a series of passageways **146** which can include a plurality of check valves **120**, a pump and control valve. Embodiments of accumulator **112**, passageways **146**, single acting cylinder **104**, and embedded magnet actuator **116** allow fluid (e.g., oil) to communicate or flow to accumulator **112** from single acting cylinder **104** and to single acting cylinder **104** from accumulator **112** in response to movement of armature **134** and the magnetic flux created by current passing through coil **130** when activated with a first and second polarity direction.

[0061] Cylinder body **102** includes a fill port **106**, which is fluidly connected to passageways **146**. Fill port **106** provides an opening into passageways **146** such that fluid and/or oil can be provided into passageways **146**. Fill port **106** is fluidly connected to passageway **146a** which is fluidly connected to check valve **120a**. Check valve **120a** is fluidly connected to pump reservoir **150**. Check valve **120a** is operable to allow the flow of fluid from passageway **146a** to pump reservoir **150** and passageways **146** throughout the assembly **100** until the fluid pressure from the side of the fill port **106** reaches a predetermined threshold. Upon removal of fluid pressure at the fill port **106**, the check valve **120a** will prevent fluid to pass from cylinder passages **146** and pump reservoir **150** into passageway **146a**. Pump reservoir **150** is fluidly connected to check valve **120c**.

[0062] Check valve **120c** is fluidly connected to pump reservoir **150** and passageway **146b** (shown in FIG. 3). Check valve **120c** is operable to prevent the flow of fluid from pump reservoir **150** to passageway **146b** until the fluid pressure from the pump reservoir **150** reaches a predetermined threshold, which is created by lowering the pressure in cavity **146b** when the pump **126** is activated by compressing spring **142** with the embedded magnet actuator rod **140** causing a high differential pressure across check valve **120c**. Passageway **146b** is fluidly connected to check valve **120d**. Check valve **120d** is fluidly connected to passageway **146c**, check valve **120b**, and drain valve **118**. Passageway **146c** is fluidly connected to relief valve **154**, which is fluidly connected to pump reservoir **150**.

[0063] Passageways **146b** and **146c** each include a bleed screw **108**. Bleed Screws **108** are operable to allow a temporary opening between passageways **146b**, **146c** and the surrounding environment such that air can be removed from passageways **146b**, **146c**.

[0064] Drain valve **118** and check valve **120b** is adjacent one distal end of rod **140**. Piston **126** with check valve **120c** is adjacent to the distal end of rod **140** that is opposite the drain valve **118**. It should be appreciated that embodiments of drain valve **118** include an electrically operated solenoid drain valve in which current can be passed through a coil in the solenoid causing an armature in the solenoid to move and either open or close the drain valve. Embodiments provide that movement of armature **134** and rod **140** in response to current passing through coil **130** activated in a first polarity direction will cause rod **140** to come into contact with piston **126** causing piston **126** with check valve **120c** to move in the longitudinal direction towards accumulator **112** until armature **134** hits the pole stop **188** (shown in FIG. 21). During this movement, fluid will be allowed to flow from pump reservoir **150** to passageway **146b** between check valve **120c** and **120d**. The amount of fluid from the reservoir that is allowed to flow will be equivalent to the increase in volume created during the movement of piston **126**. The check valve **120c** will close when the volume of fluid displaced is equal to the volume of movement of piston

126. When current is removed from coil **130**, the piston **126**, rod **140**, and armature **134** will be pushed back towards their neutral positions by spring **142** until piston **126** stops on retainer **182** (shown in FIG. **21**), which provides a lip or physical stop for piston **126**. The motion of the armature **134** when current is removed from coil **130** creates fluid pressure that opens check valve **120d** and opens a connection into actuator cavity **146c**. The amount of fluid moved is equal to the amount of fluid displaced by piston **126** and check valve **120c** when the armature **134** moves away from the accumulator along the longitudinal axis **110** until the piston **126** contacts the retainer **182** and movement is stopped. The pressure from the fluid will then act upon single acting cylinder **104** and will cause single acting cylinder **104** to move in the longitudinal direction against spring **160**. The amount of movement of single acting cylinder **104** is based on the amount of fluid displaced per cycle of pump piston **126** and closed check valve **120c** during the deactivation cycle of coil **130**.

[0065] Embodiments provide that movement of armature **134** and rod **140** in response to current passing through coil **130** in a second polarity direction will cause rod **140** to come into contact with check valve **120b** and drain valve **118** and will cause check valve **120b** and drain valve **118** to move in the longitudinal direction to the open position. In the open position, fluid built up from accumulator **112** acting against single acting cylinder **104** will be allowed to flow through drain valve **118** from passageway **146c** through passageway **146f** toward accumulator reservoir **150**. Relief valve **154** is fluidly connected to passageway **146c** and pump reservoir **150**. Relief valve **154** is operable to allow a flow of fluid from passageway **146c** to pump reservoir **150** when the pressure from the fluid within passageway **146c** reaches a predetermined threshold at which point relief valve **154** will allow a flow of fluid until the pressure within passageway **146c** reaches a certain threshold.

[0066] Embodiments of device **100** further include an optional position sensor **160** (shown in FIG. **4**). Position sensor **160** is operably connected to single acting cylinder **104** and is able to sense a relative location of single acting cylinder **104** in the longitudinal direction. As depicted in FIG. **4**, embedded magnet actuator **116** is operated by a controller **162** that is operable to cause a current to pass through coil **130**, which will cause armature **134** to move in the longitudinal direction. Embodiments of controller **162** include an electronic controller with an H-bridge that can be built into the controller or can be a separate and remote from the controller. As shown in FIG. **4**, the current that passes through coil **130** will cause the polarity of the coil **130** to change to and from $-/+$ (causing the armature **134** with rod **140** to move in the longitudinal direction toward the accumulator **112**) to $+/-$ (causing the armature **134** with rod **140** to move in the longitudinal direction toward the drain valve **118**). It should be noted that the polarity that is applied to coil **130** to cause directional motion of armature **134** is dependent on the orientation of the magnets **132** with respect to one another as installed in core tube **138** as well as the direction of the winding of coil **130**. Device **100** also includes vents or ports that are operable to move between an open position and a closed position and are fluidly connected to passageways **146** to allow air to be removed from passageways **146**.

[0067] Reference is now made to FIG. **5**, which illustrates a perspective view of device **100**. In FIG. **5**, the exterior wall of device **100** is transparent such that the interior elements of device **100** are visible. Also shown in FIG. **5** is position sensor **160**, which is operably coupled to device **100**. In one embodiment, device **100** has a diameter of 2.75 inches, a flange diameter of 3.25 inches and a length of 7.08 inches. It should be appreciated that embodiments of device **100** include any size provided it operates as set forth herein.

[0068] Illustrated in FIG. **6** is an exemplary flow path of fluid that passes through passageways **146** when embedded magnet actuator **116** is caused move towards pump A and accumulator **112**. In this embodiment, the arrows within passageways **146** illustrate that fluid will drain from accumulator **112** and move through pump A toward to passageway **146** toward cylinder **104** and cavity A. In should be appreciated that in the embodiments depicted in FIGS. **5-6**, **9-11**, and **13-20**, cylinder **104**

is a double acting cylinder (as opposed to the single acting cylinder **104** as shown in FIGS. 2-3). Embodiments of the double acting cylinder and the single acting cylinder include double acting actuators and single acting actuators, respectively. It should be appreciated that embodiments of the double acting cylinder and the single acting cylinder are not limited to cylinders, but include vanes and actuators that are not cylinders. Here, cylinder **104** includes a cavity A and a cavity B that are each operable to maintain a fluid and have such fluid moved into and out of such cavity. It should also be appreciated that due to the arrangement of rod **140** with respect to the pump pistons **168**, **126** and check valves **120c**, **172** disposed in pumps A and B as depicted in FIGS. 5-7, 9-15, and 21-25, the embedded magnet actuator **116** is operable to charge the pump pistons **168**, **126** with fluid when it compresses the pistons away from the neutral position while their respective springs are operable to pump fluid through the system upon return to their neutral or centered position. FIG. 7 illustrates a close-up view of pump A. Pump A is adjacent to rod **140** and rod pin **164**. Pump A piston assembly **166** that includes a spring loaded check valve having a pump piston **168**, ball holder **170**, a ball **172**, a first spring **174** and a second spring **176**. Ball **172** is maintained within port **178** of pump piston **168** by first spring **174** obstructing a flow of fluid therethrough. The location of pump piston **168** is maintained by spring **176**. When rod **140** with rod pin **164** moves towards pump piston **168**, second spring **176** is compressed and the fluid is allowed to flow. In one embodiment, pump A piston assembly **166** can be replaced with a shuttle valve pump piston assembly **178** (shown in FIG. 8a) or pump piston assembly **178** having a reed valve **180** (shown in FIG. 8b). Rod pin **164** slidably disposed within body **500** (shown in FIG. 7) and isolates the fluid within core tube **138** from fluid within pump A by preventing fluid from passing between the interior of the core tube **138** and pump A. It should be appreciated that the portion of rod **140** adjacent pump B does not include a rod pin. As such, fluid is able to flow between the core tube **138** and pump B.

[0069] Reference is now made to FIGS. 9-15, which illustrate an exemplary device **100** having two drain valves **152a**, **152b**. In this embodiment, cavity A of double acting cylinder **104** is fluidly connected to drain valve **152a** via passageway **146d**. Similarly, cavity B is fluidly connected to drain valve **152b** via passageway **146e**. Drain valves **152a**, **152b** are operable to allow a flow of fluid from cavities A and B to accumulator reservoir **150**. FIG. 10 illustrates a cross-sectional view of device **100** having two drain valves **152a**, **152b** illustrating passageway **146f** from pump B to cavity B. In this embodiment, passageway **146f** includes a check valve **120e**. FIG. 11 illustrates a cross-sectional view of device **100** having two drain valves **152a**, **152b** with passageways **146g** from accumulator **112** to an inlet of pump B. Referring to FIG. 12, shown is device **100** having two drain valves **152a**, **152b** illustrating passageway **146h** from cavity B to drain valve **152a**. Here, a pilot piston will open the drain valve **152a** to drain cavity A of double acting actuator from passage **146d** to the accumulator reservoir **150** in response to the fluid pressure in cavity B.

[0070] Reference is now made to FIG. 13, which illustrates an exemplary block diagram of device **100**. It should be appreciated that the lines connecting the block elements illustrated in the block diagrams depicted herein represent the passageways that allow fluid to flow therebetween and therefore fluidly connect such block elements. In this embodiment, device **100** is operable to charge the single coil embedded magnet solenoid and move an armature either towards pump A or pump B to allow a flow of fluid toward the other pump that has not been compressed. In this embodiment, device **100** includes relief valve A and relief valve B. Relief valve B is fluidly connected to the accumulator and the inlet of Pump B and can drain fluid from cavity B of the double acting cylinder. Relief valve A is fluidly connected to the accumulator and the inlet of pump A and can drain fluid from cavity A of the double acting cylinder. Device **100** shown in FIG. 14 operates similarly that shown in FIG. 13, however, there are no relief valves in this embodiment. In the embodiment shown in FIGS. 4, 13-15, and 22-25 each pump limits the maximum pressure by using the spring for the pumping. The mechanical limitations of the spring limit the output pressure capability of the pump. FIG. 15 illustrates another exemplary device **100**. In this embodiment,

pumps A and B include valves **1502**, **1504** rather than a spring loaded check valve. [0071] Referring to FIGS. **16-20**, shown are exemplary steps in which the embedded magnet actuator **116** will cause fluid to flow through device **100**. The arrows in the passageways in FIGS. **16-20** illustrate the direction of flow of fluid through the paths. It should be appreciated that in the embodiments depicted in FIGS. **15-20** and **26-27**, the embedded magnet actuator **116** is operable to pump fluid through the system by compressing the pistons **1602**, **1604** (movement away from neutral) while the springs **1603**, **1605** within the pistons **1602**, **1604** will cause the pistons **1602**, **1604** to charge with the fluid when the rod **1606** and armature **1608** from the embedded magnet actuator **116** return to their neutral or centered position. In FIG. **16**, the spring **1603**, **1605** within the piston **1602**, **1604** causes the piston **1602**, **1604** to return opening check valve **1610** and charge the pump with fluid or oil. At this point, the check valve **1610** in pump **1604** opens and check valve **1612** will be closed obstructing the flow of fluid to the double acting actuator. When the coil **130** is energized with a first polarity, the embedded magnet actuator **116** pushes the piston of pump **1604** towards the accumulator **112** causing check valve **1610** in pump **1604** to close and check valve **1612** to open which allows the volume of fluid being pumped to flow through check valve **1612** and into the double acting actuator cavity A. The force applied by embedded magnet actuator **116** minus the spring load divided by the pump piston area is the pressure applied to the double acting actuator piston. In FIG. **17**, embedded magnet actuator **116** pumps fluid through the system and compresses the springs. Embedded magnet actuator **116** is operable to compress pump B when the coil is operated with a second polarity to allow a flow of fluid through a passageway in communication to double acting actuator cavity B, inlet of drain valve B, and pilot piston of drain valve A (not shown). Also shown is the fluid communication from the accumulator reservoir to inlets of pump A and pump B. The pilot pressure from pump A shown, acts on the piston and opens drain valve B to allow cavity B to drain when pump A is operating and pump B is not. Also shown is the fluid bleed from cavity B to the environment.

[0072] In FIG. **18**, the embedded magnet actuator **116** is operable to compress pump A to allow a flow of fluid to cavity A. Relief valve A is operable to allow a flow to accumulator **112** if the pressure from the fluid reaches a certain threshold. In FIG. **19**, embedded magnet actuator **116** compresses pump B and applies pressure to the pilot piston of drain valve A. This allows a flow of fluid from cavity A to flow through the passageways from double acting actuator cavity A to drain valve A and to the accumulator reservoir fluidly connected at the inlet of pump A. Referring to FIG. **20**, shown is an exemplary device **100** having check valves **2002**, **2004** intermediate pump A, pump B and relief valve A and B, respectively.

[0073] Referring to FIG. **22**, shown is a block diagram of an alternative embodiment of device **100**. In the embodiment shown in FIG. **22**, device **100** includes drain valve A which is fluidly connected to the passageway flowing from the embedded magnet solenoid to check valve **3**, and drain valve B which is fluidly connected to the passageway flowing from the embedded magnet solenoid to check valve **2**. This arrangement allows the double acting cylinder to remain in a static position when the embedded magnet solenoid is powered off. Also shown in FIG. **22** are two controlled leaks. Each controlled leak is located past the pilot pistons of drain valve A and B and are fluidly connected to the accumulator reservoir. Embodiments provide that the controlled leaks can allow a flow of fluid at a plurality of different rates. FIG. **22** also depicts two optional check valves. Optional check valves are located in series with the pilot pistons of drain valves A and B, respectively. It should be noted that the optional check valve can be deployed within or integral with the drain valves A and B rather than as a separate element. Referring to FIG. **23**, shown is another block diagram of an embodiment of device **100**. In this embodiment, device **100** includes a safety drain valve **2300**. This arrangement also allows the cylinder to remain static when the embedded magnet solenoid is powered off. The safety drain valve **2300** allows for the fluid to be drained from cavity A if the system loses power allowing the cylinder to move towards cavity A that the oil is being drained from. It should be appreciated that the location of the safety drain valve **2300** with respect to the

other elements of device **100** is determined by the fail safe direction required of the double acting cylinder. Embodiments provide that the safety drain valve **2300** can be disposed such that it is operable to drain fluid from cavity B instead of cavity A as depicted in FIG. **23** allowing the cylinder to move towards cavity B that the oil is being drained from. FIG. **23** also illustrates two controlled leaks. In this embodiment, controlled leaks are located past the pilot pistons of drain valves A and B, respectively, and are located in parallel to safety drain valve **2300**. Also, shown in FIG. **23** are optional check valves disposed in series with the pilot piston of drain valves A and B, respectively. It should be appreciated that the optional check valve and controlled leaks can be disposed within or integral to Drain Valves A and B, and that the controlled leaks can be located within the pilot piston of drain valves A and B. Reference is now made to FIG. **24**, which depicts yet another block diagram of an embodiment of device **100**. In this embodiment, the relief valves have been removed from device **100**, which limits the maximum fluid pressure in the system. As depicted, pump A and pump B are arranged such that movement of the armature of the embedded magnet solenoid toward either pump A or pump B compresses the spring within their respective pistons. After the pump pistons are compressed and the armature moves back to its center or neutral position, the springs cause the pump pistons to move back to their neutral position thereby pumping fluid through the passageways. Since the springs are causing or forcing the fluid to flow, the upper limits of the force the springs are able to generate act as a cap or maximum for the amount of pressure that can be generated within the device **100**. In FIG. **24**, controlled leaks are shown as being located past the pilot piston to the accumulator reservoir. Controlled leaks are disposed in series with optional check valves, which are positioned such that they are in parallel with drain valve pilot piston A and B. It should be appreciated that the controlled leaks can be disposed with the pilot piston of drain valves A and B.

[0074] Referring now to FIG. **25**, depicted is another block diagram of yet a further alternative embodiment of device **100**. This embodiment is similar to that found in FIG. **24**, however, in this embodiment, device **100** includes a safety drain valve **2300** in addition to having drain valves A and B. It should be appreciated again that the location of the safety drain valve **2300** is determined by the fail safe direction required of the double acting cylinder towards the cylinder the oil is being drained from by the safety drain valve. Embodiments of the safety drain valve **2300** include it being located within device **100** such that it is operable to drain fluid from cavity B instead of cavity A as depicted in FIG. **25**. Similar to FIG. **24**, FIG. **25** includes controlled leaks being located past the pilot piston to the accumulator reservoir in parallel with optional check valves. Optional check valves illustrated in FIG. **24** are located in parallel to drain valve pilot piston A and B, respectively. It should be noted that the optional check valve and the controlled leaks can be deployed within drain valves A and B. The controlled leaks can be disposed within the pilot piston of drain valves A and B. Referring to FIG. **26**, shown is another block diagram of another embodiment of device **100**. In this embodiment, device **100** includes check valves 2 and 5. Check valve 2 is disposed between pump B and cavity B of the cylinder and its outlet is connected to relief valve B inlet and cylinder cavity B. Check valve 5 is disposed between pump A and cavity A of the cylinder and its outlet is connected to relief valve A inlet and cylinder cavity A. As indicated by the dotted line in FIG. **26**, Drain valve A pilot section is connected between check valves 1 and 2. As indicated by the dotted line in FIG. **26**, Drain valve B pilot section is connected between check valves 4 and 5. FIG. **26** also illustrates controlled leaks located past the pilot piston to the accumulator reservoir. In this embodiment, controlled leaks are disposed in series with optional check valves. Optional check valves are disposed in series with the pilot piston of drain valves A and B, respectively. It should be noted that the controlled leak and optional check valves can be located within the drain valves A and B. The controlled leaks can be deployed within pilot pistons A and B. FIG. **27** presents a block diagram of device **100** similar to that found in FIG. **26**. However, in this embodiment, there is a safety drain valve **2300** disposed in parallel with the drain valve A. It should be appreciated that the location of safety drain valve **2300** is determined by the

fail safe direction required of the double acting cylinder. Embodiments provide that the safety drain valve **2300** can be located such that it is operable to drain fluid from cavity B when connected in parallel to drain valve B. As shown, drain valve A pilot is fluidly connected between check valves 1 and 2, and drain valve B pilot is connected between check valves 4 and 5. FIG. 27 illustrates controlled leaks located past the pilot piston to accumulator reservoir and are in series with optional check valves. Optional check valves are disposed in series with the pilot piston of drain valves A and B. It should be appreciated that the controlled leaks can be disposed within the drain valve pilot piston. It should also be noted that the optional check valve can be located within drain valves A and B. One of the controlled leaks is located in parallel with safety drain valve **2300**. It should be appreciated that in each of the embodiments of device **100** illustrated in FIGS. 22-27, the double acting cylinder is operable to remain in a static position when the embedded magnet solenoid is powered off.

[0075] It should be appreciated that FIGS. 28-30 show additional embodiments of device **100** in which pilot operated logic control valves that control the motion of the embedded magnet actuator when it receives a pressure signal from the solenoid pumping system as previously described herein to the control valves as described and shown herein. FIG. 27 illustrates an embodiment of device **100** that utilizes two three-way pilot operated valves, which are typically in the closed position. FIG. 29 shows an embodiment that utilizes two three-way pilot operated valves that are typically in the open position. FIG. 30 shows one 3-position, 5-ported, spring centered pilot operated valve.

[0076] Reference is now made to FIGS. 31 and 32, which depict block diagrams of exemplary electro-hydrostatic actuator assembly **3100** having a first accumulator **3108** and a second accumulator **3110**. Shown in FIGS. 31 and 32 are electro-hydrostatic actuator assembly **3100** having a cylinder body **3102** (shown in FIG. 33). Cylinder body **3102** includes a single acting cylinder **3104** (shown in FIG. 31) and a fill port **3106**. Alternatively, cylinder body **3102** includes a double acting cylinder **3204** (shown in FIG. 32). Electro-hydrostatic actuator assembly **3100** also includes a first accumulator **3108**, a second accumulator **3110**, a first piston pump **3112**, a second piston pump **3114**, an embedded magnet actuator **3116**, a pressure isolation pin **3118**, a first check valve **3120**, a second check valve **3122**, a release valve **3124**, a 3-way pilot operated valve **3126** and an air bleed valve **3128**. Embodiments include orifices **3130**, **3132**. It should be appreciated that single acting cylinder **3104** includes a position sensor **3105** operable to sense a position of the location of the single acting cylinder **3104**.

[0077] Embedded magnet actuator **3116** is similar to that embedded magnet actuator **3116** described above. However, in this embodiment, embedded magnet actuator **3116** is operable to activate first piston pump **3112** and second piston pump **3114**. In this embodiment, passing current through its coil **3134** of embedded magnet actuator **3116** in a first polarity direction will cause armature **3136** to move toward the first piston pump **3112** in a first direction. This will cause the first check valve within first piston pump **3112** to open and allow a flow of fluid through it. Movement of armature **3136** toward first piston pump **3112** in the first direction will also cause pressure isolation pin **3118** to compress first pump spring **3138** and will draw or urge fluid from the first accumulator **3108**.

[0078] When current through coil **3134** stops, armature **3136** will return to a centered or neutral position within embedded magnet actuator **3116** such that it is not acting on either first piston pump **3112** or second piston pump **3114**. When armature **3136** is moving toward the neutral position the first pump spring **3138** of the first piston pump **3112** will push or move the first piston pump **3112** in a second direction opposite the first direction towards the second piston pump **3114**, and the first check valve **3120** will open allowing a flow of fluid. Movement of the first piston pump **3112** by moving the first pump spring **3138** from the compressed state to an uncompressed state will urge or move fluid in the first piston pump **3112** through the second check valve **3120** into the second accumulator **3110**. After the volume of fluid from the first piston pump **3112** has moved to the second accumulator **3110**, check valve **3120** will close preventing a flow of fluid.

[0079] The second accumulator **3110** is fluidly connected to the second piston pump **3114** via an inlet of the second piston pump **3114** by one of a plurality of passageways within the cylinder body **3102**. When current is passed through its coil **3134** of embedded magnet actuator **3116** in a second polarity direction will cause armature **3136** to move toward the second piston pump **3114** in a second direction. The second polarity is opposite the first polarity. The movement of the armature **3136** in the second direction is opposite the first direction. Movement of the armature **3136** to move toward the second piston pump **3114** in a second direction will cause the third check valve **3140** to open and allow a flow of fluid through it. Movement of armature **3136** toward second piston pump **3114** in the second direction will also cause the rod of embedded magnet actuator **3116** to compress second pump spring **3142**. This will cause second piston pump **3114** to draw or urge fluid from the second accumulator **3110** via the fluidly connected passageway. When the current ceases to pass through coil **3134** of embedded magnet actuator **3116**, armature **3136** will move to the neutral position. The neutral position pertains to a location along the long axis of the embedded magnet actuator **3116** in which armature **3136** is centered and the rod is not compressing both the first piston pump **3112** and second piston pump **3114**. The second pump spring **3142** will then move from a compressed state to an uncompressed state, the third check valve **3140** will close. Next, a fourth check valve **3144** will open and allow pressurized flow of fluid into the single acting cylinder **3104**. The flow of fluid into the single acting cylinder **3104** each time the second pump spring **3142** is moved from a compressed state to an uncompressed state will correspond to the volume of fluid maintained by the second piston pump **3114** when the second pump spring **3142** is in the compressed state. Accordingly, fluid will stop flowing into the single acting cylinder **3104** when the second pump spring **3142** is in the uncompressed state.

[0080] It should be appreciated that embodiments include the fluid pressure created by the first piston pump **3114** when fluid is moved by the first piston pump **3114** from the first accumulator **3108** to the second accumulator **3108** will be at a first level. The fluid pressure created by the second piston pump **3114** when fluid is moved by the second piston pump **3114** from the second accumulator **3108** to the single acting cylinder **3104** will be at a second level. Embodiments include the second level being greater than the first level. In this regard embodiments of the electro-hydrostatic actuator assembly **3100** are operable to create higher levels of fluid pressure than an assembly having only a single accumulator.

[0081] Referring to FIGS. **32-35**, shown is an alternative embodiment of an electro-hydrostatic actuator assembly **3200** having a cylinder body **3202**. Electro-hydrostatic actuator assembly **3200** includes a first accumulator **3208**, a second accumulator **3210**, a first piston pump **3212**, a second piston pump **3214**, an embedded magnet actuator **3216**, a pressure isolation pin **3218**, a first check valve **3220**, a second check valve **3222**, a third check valve **3223**, a release valve **3224**, first air bleed valve **3228**, and a second air bleed valve **3229**. In this embodiment, movement of armature **3236** towards the first piston pump **3212** causes first piston pump **3212** to move fluid from the first piston pump **3212** to the second accumulator **3210** and check valve **3222** to open. The fluid will then move to cavity A **3223** of double acting cylinder **3204**. When armature **3236** moves back to a neutral position, first piston pump **3212** is urged by first pump spring **3238** to return to an uncompressed state, which moves fluid into first piston pump **3212** from the first accumulator **3208**. When the armature **3236** moves towards the second piston pump **3214**, the pressure isolation pin causes the second piston pump **3214** to move fluid from the second accumulator **3210** to cavity B **3225** of the double acting hydraulic cylinder **3204**. When the armature **3236** moves back to the neutral position, second piston pump **3214** is urged by second pump spring **3242** to the uncompressed state, which moves fluid into second piston pump **3214** from the second accumulator **3210**.

[0082] Referring to FIGS. **36-41**, shown are exemplary embodiments of a flexible reed valve **4300**. An embodiment of reed valve **4300** is also illustrated as reed valve **180** in FIG. **8b**. Embodiments of flexible reed valve **4300** can be utilized in hydraulic actuators depicted herein, and any type of

fluid pumps or fluid pump systems. Embodiments of flexible reed valve **4300** can be used in place of check valves in multiple locations in hydraulic systems and pneumatic systems. Embodiments of a flexible reed valve **4300** include an inlet that allows fluid to flow into the flexible reed valve **4300**, and an outlet that allows fluid to flow out of the flexible reed valve **4300**. Embodiments of flexible reed valve **4300** are operable to obstruct and prevent the movement or flow of a fluid through a fluid pump or fluid pump system until a threshold pressure (also known as a cracking pressure) is reached on the inlet side of the flexible reed valve **4300**. Once the threshold pressure differential (or cracking pressure) is reached between the inlet side and the outlet side of the flexible reed valve **4300**, the flexible reed valve **4300** flexes or bends allowing the fluid on the inlet to flow through the flexible reed valve **4300** and out the outlet of the flexible reed valve **4300**. If the pressure differential is below the threshold pressure differential then the valve will close and prevent a flow of fluid. If the pressure differential (i.e., the difference between the fluid pressure on the inlet side and the outlet side) of fluid is below the threshold pressure differential, the flexible reed valve **4300** will not allow fluid to flow from the inlet side to the outlet side. In this regard, if the flexible reed valve **4300** allows fluid to flow from the inlet and out the outlet, the pressure from fluid on the inlet side of the flexible reed valve **4300** may decrease below the threshold pressure. Once the threshold pressure on the inlet side of the flexible reed valve **4300** falls below the threshold pressure, the flexible reed valve **4300** will flex or bend in an opposite direction to that which allowed the fluid to flow such that the flexible reed valve **4300** no longer allows the fluid to flow from the inlet to the outlet thereby preventing the flow of fluid through it. In other words, embodiments of the flexible reed valve **4300** are operable to reseal and not allow a flow of fluid once the pressure from the fluid on the inlet side of the flexible reed valve **4300** falls below the threshold pressure.

[0083] Referring to FIG. 36, shown is flexible reed valve **4300** located within a pump piston **4302**. Flexible reed valve **4300** includes a fixed body portion **4304** and a flexible body portion **4306**. Fixed body portion **4304** is located adjacent flexible body portion **4306**. Fixed body portion **4304** includes an outer lip that circumscribes the main body portion **4317** of the fixed body portion **4304** and maintains a location of fixed body portion **4304** and flexible body portion **4306** with respect to pump piston **4302**. As shown in FIG. 36, fixed body portion **4304** includes a first side **4301**, an opposite second side **4303**, and an opening **4308**. Opening **4308** fluidly connects the side of fixed body portion **4304** opposite the flexible body portion **4306** with the side of fixed body portion **4304** adjacent the flexible body portion **4306**. Flexible body portion **4304** (as shown in FIG. 37) includes a flexing portion **4310** and a stationary portion **4312**. Stationary portion **4312** is fixedly attached to fixed body portion **4304** and does not flex or bend with respect to fixed body portion **4304**. In other embodiments stationary portion **4312** is not fixedly attached to the fixed body portion **4304**, but it is maintained in place within the passageway through a friction fit. As shown in FIG. 37, stationary portion **4312** is fixedly attached to fixed body portion **4304** by snap fittings **4314**. It should be appreciated that embodiments of stationary portion **4312** can be fixedly attached to fixed body portion **4304** by any means that does not cause fluid to leak or pass through flexible body portion **4306**.

[0084] Flexible body portion **4304** includes a reed or flap **4316**. The flexible body portion **4304** include a first side **4305** and an opposite second side **4307**. Flap **4316** is located approximately in the center of flexible body portion **4304**. Flap **4316** is sized such that it is at least the same or greater size as that of opening **4308** such that flap **4316** obstructs the flow of fluid through opening **4308** (shown in FIG. 38). The location of flap **4316** coincides with that of opening **4308** such that flap **4316** completely overlaps with opening **4308**. Flap **4316** is spaced from stationary portion **4313** on three sides **4318** and is connected to stationary portion **4312** along a fourth side **4320**. Embodiments provide that flap **4316** is integral with stationary portion **4312** via its connection with side **4320**. It should be appreciated that embodiments of provide that flap **4316** take any shape provided that it is connected to stationary portion **4313** in a fashion that allows flap **4316** to bend or

flex such that flap **4316** is operable to move away from opening **4308**. Flexing or bending of flap **4316** creates at least one passageway **4309** that fluidly connects the first side **4305** and the second side **4307**.

[0085] In practice, flexible reed valve **4300** is disposed within a pump piston **4302**. Pump piston **4302**, in the embodiment shown in FIG. **43**, is operable increase fluid pressure on the side of flexible reed valve **4300** having the fixed body portion **4304**. Once the fluid pressure differential between the fluid on the fixed body portion **4304** and the flexible body portion **4306** reaches a predetermined threshold, the flap **4316** will flex or bend such that flap **4316** moves away from the opening **4308** and fluid is allowed to flow through opening **4308**. Once the fluid pressure differential between the fluid on the fixed body portion **4304** and the flexible body portion **4306** falls below the predetermined threshold, the flap **4316** will bend or flex back towards the fixed body portion **4304** obstructing the flow of fluid through opening **4308**.

[0086] Reference is now made to FIG. **39**, which illustrates another embodiment of a flexible body portion **4606** of a flexible reed valve **4600**. Shown in FIG. **39** is flexible body portion **4606**. Flexible body portion **4606** includes a body **4602** and a plurality of openings **4604**. The plurality of openings **4604** provide a passage way from one side of the body **4602** to the other side of the body **4602**. In this embodiment, flexible body portion **4606** does not have a single piece or flap that bends or flexes to obstruct fluid flow or allow fluid flow. Rather, flexible body portion **4606** bends or flexes along each of the fingers **4608** that are spaced from one another by the plurality of openings **4604**, which allows center portion **4610** to move away from the opening of that is obstructing the flow of fluid. As shown in FIG. **39**, flexible body portion **4606** includes three fingers **4608** that form a spiral design.

[0087] Shown in FIGS. **40** and **41** are flexible body portion **4606** located within a piston pump **4700**. Flexible body portion **4606** is located adjacent to fixed body portion **4602**. As shown in FIG. **41**, fixed body portion **4612** includes a plurality of openings **4614** that are disposed adjacent to the plurality of openings **4604**. Center portion **4610** is sized to be at least the same size or greater than the passageway **4702** of piston pump **4700**. In this embodiment, center portion **4610** obstructs a flow of fluid from passageway **4702** until the pressure differential between the fluid in passageway **4702** and fluid on the other side of flexible body portion **4602** reaches a predetermined threshold. Once the pressure differential reaches the predetermined threshold, center portion **4610** will move towards fixed body portion **4612** and away from passageway **4702** such that passageway **4702** is not obstructed and fluid can flow through the plurality of openings **4604** and plurality of openings **4610**. The center portion **4610** of flexible body portion **4602** is operable to move with respect to the outer portion **4616** of flexible body portion **4602** by the flexing or bending of the fingers **4604**. It should be appreciated that embodiments of the flexible reed valve described herein can be used in place of any of the check valves shown or described herein. Accordingly, embodiments of the flexible reed valve are operable to be used in a fashion in which an embedded magnet actuator can actuate flexible reed valve such that it either (1) causes (i.e., pumps) fluid to move through the system, and the flexible reed valve is charged with fluid when the embedded magnet actuator returns to neutral, or (2) causes (i.e., charges) fluid to move into a piston pump and flexible reed valve causes (i.e., pumps) fluid to move through the system.

[0088] It is to be understood that any feature described in relation to any one embodiment may be used alone, or in combination with other features described, and may also be used alone, or in combination with one or more features of any other of the embodiments, or any combination of any other of the embodiments. The presently disclosed embodiments are therefore considered in all respects to be illustrative. Furthermore, equivalents and modifications not described above may also be employed without departing from the scope of this disclosure, which is defined in the accompanying claims

Claims

1. An actuator assembly comprising: a cylinder body having a plurality of passageways; an accumulator fluidly connected to the plurality of passageways, the accumulator operable to maintain a fluid under pressure and to urge the fluid through the plurality of passageways; a moveable cylinder fluidly connected to the plurality of passageways, the moveable cylinder operable to move in response to pressure from the fluid; and a solenoid assembly comprising a core tube, an armature, a first magnet, a second magnet, and an excitation coil, the core tube extending through a longitudinal axis, the core tube having a first ferromagnetic end section and a spaced apart second ferromagnetic end section, the armature having a rod extending through the longitudinal axis, the armature disposed longitudinally between the first ferromagnetic end section and the spaced apart second ferromagnetic end section, the first magnet and the second magnet disposed radially outside the core tube, the first magnet separated from the second magnet by a ferromagnetic spacer, the excitation coil located radially outward from the first and second magnets.
2. The actuator assembly according to claim 1, wherein the plurality of passageways fluidly connect the accumulator, the solenoid assembly and the moveable cylinder.
3. The actuator assembly according to claim 2, wherein the accumulator is fluidly connected to the solenoid assembly by a piston, and wherein the moveable cylinder having a drain valve is fluidly connected to the solenoid assembly, wherein the piston is operable to move between an open position and a closed position, and wherein the drain valve is operable to move between a drain open position and a drain closed position.
4. The actuator assembly according to claim 3, wherein the solenoid assembly is operable to move the armature with the rod through the longitudinal axis in response to a current passing through the excitation coil, wherein movement of the armature in one direction will cause the piston to move to the open position, and wherein movement of the armature in a second direction will cause the drain valve to move to the drain open position.
5. The actuator assembly according to claim 4, wherein proximal poles of the first magnet and the second magnet are a like.
6. The actuator assembly according to claim 1, wherein fluid within the solenoid assembly is isolated from fluid within at least one of (i) the accumulator and (ii) the movable cylinder.
7. The actuator assembly according to claim 6, the actuator assembly further comprising at a first flexible bearing fixedly disposed adjacent a first distal end of the rod, and a second flexible bearing fixedly disposed adjacent a second distal end of the rod, wherein the first and second flexible bearing are operable to prevent the armature and rod from contacting the core tube.
8. The actuator assembly according to claim 7, the solenoid assembly further comprising a pressure isolation pin disposed at the first distal end of the rod, the pressure isolation pin operable to prevent pressure from within the core tube from exiting the core tube adjacent the first distal end of the rod.
9. The actuator assembly according to claim 8, the actuator assembly further comprising a check valve disposed in series with the drain valve.
10. The actuator assembly according to claim 9, wherein fluid is prevented from flowing between the solenoid assembly and the cylinder when the solenoid is powered off.
11. The actuator assembly according to claim 10, wherein the moveable cylinder is a double acting cylinder have a first fluid cavity and a separate second fluid cavity, wherein fluid in the first fluid cavity and the second fluid cavity is restricted from flowing from the accumulator.
12. The actuator assembly according to claim 11, wherein the drain valve has a pilot pressure that is fluidly connected downstream from the piston and the check valve, and is upstream from a second check valve that is disposed in series with the moveable cylinder.
13. The actuator assembly according to claim 11, wherein the check valve is disposed in series with

the drain valve, wherein the drain valve is disposed fluidly connected downstream from the piston and the check valve, and the drain valve is disposed upstream of a second check valve such that fluid is prevented from flowing from the pilot piston of the drain valve back to the pump during a suction cycle of the pump.

14. An actuator assembly comprising: an accumulator fluidly connected to plurality of passageways, the accumulator operable to maintain a fluid under pressure and to urge the fluid through the plurality of passageways; a moveable cylinder fluidly connected to the plurality of passageways, the moveable cylinder operable to move in response to pressure from the fluid; and a solenoid assembly comprising a core tube, an armature, a first magnet, a second magnet, and an excitation coil, the core tube extending through a longitudinal axis, the core tube having a first ferromagnetic end section and a spaced apart second ferromagnetic end section, the armature having a rod extending through the longitudinal axis, the armature disposed longitudinally between the first ferromagnetic end section and the spaced apart second ferromagnetic end section, the first magnet and the second magnet disposed radially outside the core tube, the first magnet separated from the second magnet by a ferromagnetic spacer, the excitation coil located radially outward from the first and second magnets.

15. An actuator assembly comprising: a solenoid assembly comprising a core tube, an armature, a first magnet, a second magnet, and an excitation coil, the core tube extending through a longitudinal axis, the core tube having a first ferromagnetic end section and a spaced apart second ferromagnetic end section, the armature having a rod extending through the longitudinal axis, the armature disposed longitudinally between the first ferromagnetic end section and the spaced apart second ferromagnetic end section, the first magnet and the second magnet disposed radially outside the core tube, the first magnet separated from the second magnet by a ferromagnetic spacer, the excitation coil located radially outward from the first and second magnets, the armature comprises a rod extending through the longitudinal axis; a first pump operably coupled to a first end of the solenoid assembly, the first pump comprising a compressible piston operable to be compressed in response to movement of the rod towards the first pump; and a second pump operably coupled to a second end of the solenoid assembly, the second pump comprising a second compressible piston operable to be compressed in response to movement of the rod towards the second pump.

16. An apparatus comprising: a fixed body having a first side and an opposite second side, the fixed body defining at least one opening, the at least one opening providing a passageway between the first side and the opposite second side; and a flexible body having a flexible first side and an opposite flexible second side, the flexible body having at least one flexible portion and a spaced apart stationary portion, the at least one flexible portion operable to flex with respect to the stationary portion such that at least one passageway from the flexible first side to the opposite flexible second side is formed when the at least one flexible portion is in a flexed position.

17. The apparatus according to claim 16, wherein the at least one passage is in fluid communication with the at least one opening.

18. The apparatus according to claim 16, wherein the flexible body and the fixed body obstruct a flow of fluid between the at least one passageway and the at least one opening when the at least one flexible portion is not flexed with respect to the stationary portion.

19. The apparatus according to claim 16, wherein the flexible body and the fixed body allow a flow of fluid between the at least one passageway and the at least one opening when the at least one flexible portion is flexed with respect to the stationary portion.

20. The apparatus according to claim 16, wherein the fixed body is fixedly attached to the stationary portion of the flexible body.

21. The apparatus according to claim 16, wherein the at least one flexible portion is operable to flex with respect to the stationary portion in response to a pressure differential between a pressure on the flexible first side and the opposite flexible second side being above a predetermined threshold.

22. The apparatus according to claim 16, wherein the at least one flexible portion is operable to not

flex with respect to the stationary portion in response to a pressure differential between a pressure on the flexible first side and the opposite flexible second side being below a predetermined threshold.

23. The apparatus according to claim 16, wherein the fixed body further comprises an outer lip that is operable to maintain a location of the fixed body within a pump.

24. An actuator assembly comprising: a cylinder body having a plurality of passageways; a first accumulator fluidly connected to the plurality of passageways, the first accumulator operable to maintain a fluid under pressure and to urge the fluid through the plurality of passageways; a second accumulator fluidly connected to the plurality of passageways, the second accumulator operable to maintain a fluid under pressure and to urge the fluid through the plurality of passageways, the second accumulator fluidly connected to the first accumulator through the plurality of passageways and operable to receive fluid from the first accumulator; a moveable cylinder fluidly connected to the plurality of passageways, the moveable cylinder operable to move in response to pressure from the fluid; and a solenoid assembly comprising a core tube, an armature, a first magnet, a second magnet, and an excitation coil, the core tube extending through a longitudinal axis, the core tube having a first ferromagnetic end section and a spaced apart second ferromagnetic end section, the armature having a rod extending through the longitudinal axis, the armature disposed longitudinally between the first ferromagnetic end section and the spaced apart second ferromagnetic end section, the first magnet and the second magnet disposed radially outside the core tube, the first magnet separated from the second magnet by a ferromagnetic spacer, the excitation coil located radially outward from the first and second magnets.

25. The actuator assembly according to claim 24, wherein the plurality of passageways fluidly connect the first accumulator, the second accumulator, the solenoid assembly, and the moveable cylinder.

26. The actuator assembly according to claim 24, wherein the first accumulator is fluidly connected to the solenoid assembly by a first piston pump, wherein the moveable cylinder is fluidly connected to the solenoid assembly by a second piston pump, and wherein the second accumulator is fluidly connected to the second piston pump.

27. The actuator assembly according to claim 26, wherein the first piston pump is operable to move between an open position allowing a flow of fluid and a closed position not allowing a flow of fluid, and wherein the second piston pump is operable to move between a second pump open position allowing a flow of fluid and a second pump closed position not allowing a flow of fluid.

28. The actuator assembly according to claim 26, wherein the solenoid assembly is operable to move the armature with the rod through the longitudinal axis in response to a current passing through the excitation coil, wherein movement of the armature in one direction will cause the first piston pump to move to the open position, and wherein movement of the armature in a second direction will cause the second piston pump to move to the second pump open position.

29. The actuator assembly according to claim 24, wherein fluid within the solenoid assembly is isolated from fluid within at least one of (i) the first accumulator, (ii) the second accumulator, and (iii) the movable cylinder.

30. The actuator assembly according to claim 24, the solenoid assembly further comprising a pressure isolation pin disposed at a distal end of the rod adjacent the first piston pump, the pressure isolation pin operable to prevent pressure from within the core tube from exiting the core tube adjacent the distal end of the rod.

31. The actuator assembly according to claim 24, wherein fluid is prevented from flowing between the solenoid assembly and the cylinder when the solenoid is powered off.

32. The actuator assembly according to claim 24, wherein the moveable cylinder is a double acting actuator have a first fluid cavity and a separate second fluid cavity, wherein fluid in the first fluid cavity and the second fluid cavity is restricted from flowing from the second accumulator.
